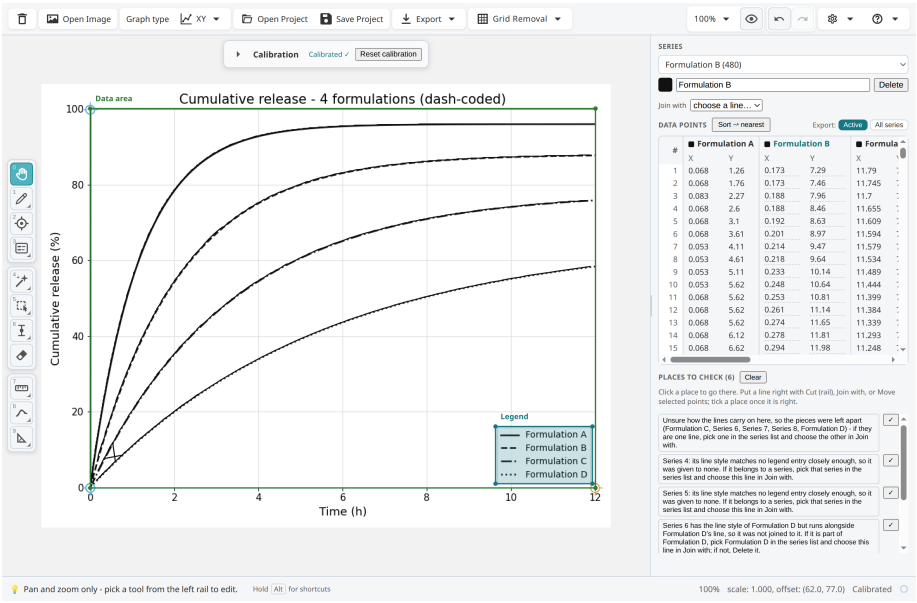
PLOTTRACER

User Guide

Turning figures back into numbers



Version 2.7 ▪ AGPL-3.0 ▪ Katalyst Nord, Stockholm

plottracer.com ▪ github.com/katalystnord/plottracer

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Chapter 1

Introduction

1.1 What PlotTracer is

An enormous amount of quantitative knowledge exists only as figures. Stress-strain curves, load-displacement data, fatigue-life plots, statistical distributions, economic time series, dose-response relationships: published in papers, reports, standards and regulatory filings, locked inside a picture, unavailable for reanalysis unless someone re-extracts the numbers. Plot digitisation is not an exotic edge case; it is a routine part of working with any published or legacy dataset.

PlotTracer is a free, fully offline, cross-platform desktop application for that job. It reads a figure, calibrates its axes, extracts the data and lets you correct and export it, across sixteen graph types, at the figure's own precision, with nothing invented and nothing hidden.

Version 2.7 brings automatic extraction into the workflow. On the figures it has been measured on, PlotTracer can find the axes, read the legend and extract every series in a few presses. Every automatic step makes the same marks a click would make, so each stays yours to check, move or redo by hand, and the two can be mixed freely. Everything runs on your machine: the text reader and the chart-element detector are bundled, and no figure ever leaves the computer.

Three principles run through every part of the tool, and are worth knowing before you open your first figure:

- **Auditable.** The code that computes your extracted coordinates is open and inspectable. A cloud service cannot be independently verified, and a proprietary binary cannot be scrutinised.
- **Available everywhere.** Linux, Windows and macOS, installed from a single package, working fully offline, including in air-gapped environments.
- **Honest about precision.** The tool records what the figure actually shows and never fabricates precision the source never carried. Where it cannot determine an answer with confidence, it says so, in the interface and in the export, rather than returning a plausible-looking number.

1.2 Who this guide is for

This guide assumes no prior experience with plot digitisers. Read it start to finish, or jump to the chart type or task you need through the table of contents. Chapter 4 is a five-minute worked example; Chapters 5 and 6 are the full reference.

1.3 Scope of this document

This guide covers the desktop application's workflow, every graph type, the analysis tools and export. It does not cover building PlotTracer from source or its internal architecture: see the project `README` for that. The shorter `MANUAL.md` in the repository, which the app's Manual button opens, covers the same ground in less space.

Chapter 2

Installing PlotTracer

PlotTracer ships as one installer per platform, built from the source repository by continuous integration. There is no account to create, and nothing is installed beyond the application itself.

Platform	Package	Notes
Linux	AppImage, `.deb`	`chmod +x` the AppImage, or `dpkg -i` the `.deb`
Windows	NSIS `.exe`	64-bit only
macOS	`.dmg`	Apple Silicon

Linux: AppImage

```
chmod +x plottracer_<version>_x86_64.AppImage
./plottracer_<version>_x86_64.AppImage
```

Linux: .deb (Debian, Ubuntu)

```
sudo dpkg -i plottracer_<version>_amd64.deb
```

The post-install step sets the sandbox permissions Chromium needs, so no manual `chown` or `chmod` is needed afterwards.

The macOS and Windows builds are currently **unsigned**. On first launch:

- **macOS**: Gatekeeper will block the app. Right-click it and choose *Open* rather than double-clicking it.
- **Windows**: SmartScreen will warn you. Click *More info*, then *Run anyway*.

Code-signing is a planned follow-up; this is expected for the current release, not a sign that anything is wrong.

Installers are attached to each tagged release on the project's GitHub page.

Chapter 3

The Interface

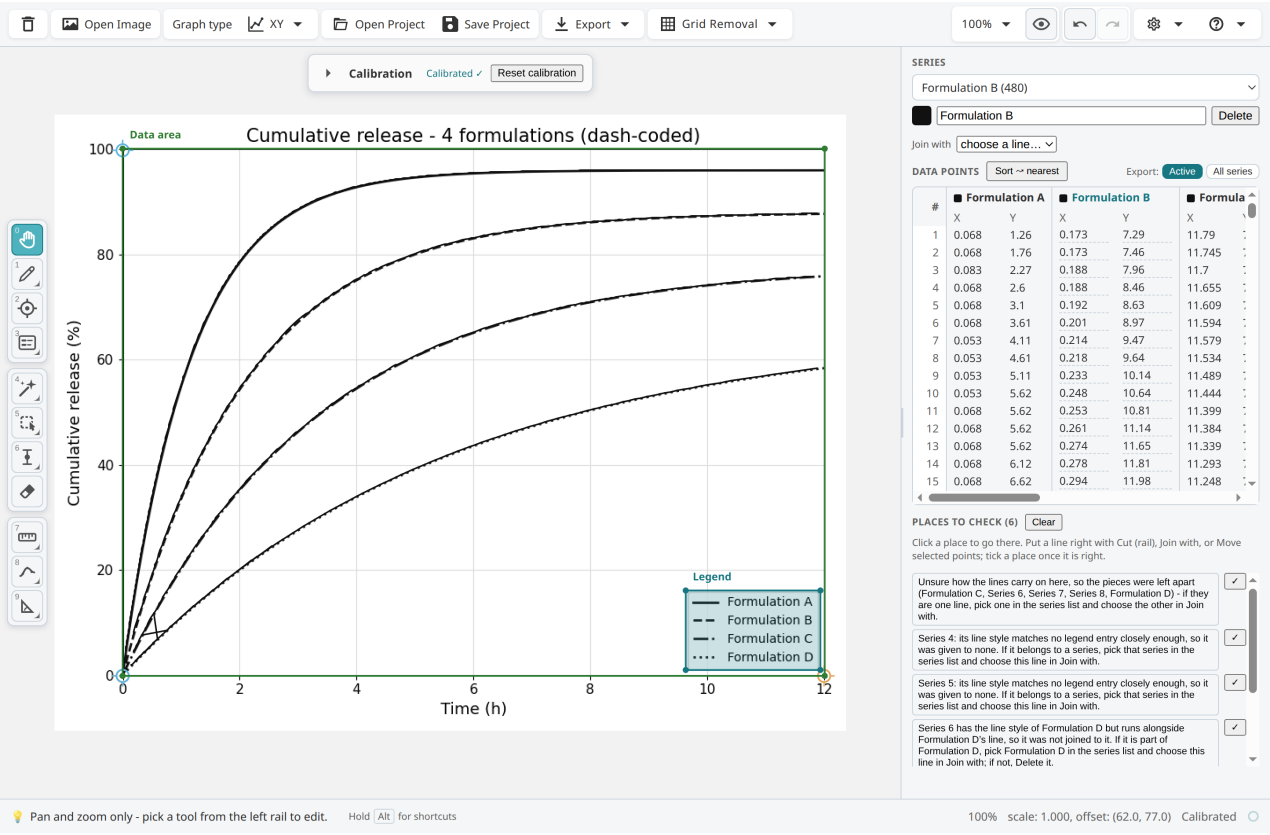


Figure 3.1: The main window: the tool rail on the left, document actions along the top, the calibration card above the figure, the extracted data on the right and the tips bar along the bottom. Here a figure with four series drawn in one ink has been read by Auto-extract, each series told apart by its line style.

Region	What lives there
Left rail	The tools, in the order of the workflow, each with its number key: `0` Pan, `1` Edit image, `2` Calibrate, `3` Legend, `4` Data extraction; then `5` Select, `6` Error bars and the Eraser; then `7` Measure, `8` Curve fit, `9` Geometry.
Top bar	Document actions: Open Image, the graph type picker, Open Project, Save Project, Export, Grid Removal, and the Help menu.
Calibration card	Above the figure once it is captured. It is always on screen, at least folded, so you can always see whether the figure is calibrated.
Canvas	The figure itself. Everything else floats above it.
Right panel	Your extracted data: one table per series, or one row per category or axis for the bar family and spider charts. This is what leaves the application when you export.
Tips bar	The bottom line says what the current tool expects next, and shows a warning triangle when something cannot proceed. The ring in its corner turns while a long operation runs.

3.1 Mouse and keyboard

Input	Action
Left button	The active tool (place, calibrate, trace, measure . . .)
`Ctrl`+Left, or middle button	Pan
Scroll wheel	Zoom
Right-click	Quick context menu (delete point, edit value, fit to view . . .)
`Enter`	Accept the current step (apply crop, finish area, run calibration)
`Esc`	Back out of the current step, or clear the selection
`Del` / `Backspace`	Delete the active point or measurement
`0` to `9`	Switch tools (mirrors the rail)
Arrow keys	Nudge the selected point or handle (`Shift`: coarse)
`Q` / `W`	Step to the previous or next point
`Ctrl+Z` / `Ctrl+Shift+Z`	Undo and redo
`F1`	The workflow and the keys, on one card

Chapter 4

Quick Start: Your First Figure

This walkthrough uses a figure bundled with every install, so you can follow along without hunting for your own first. It takes about five minutes, and every step below was driven in the application to make the pictures in this chapter.

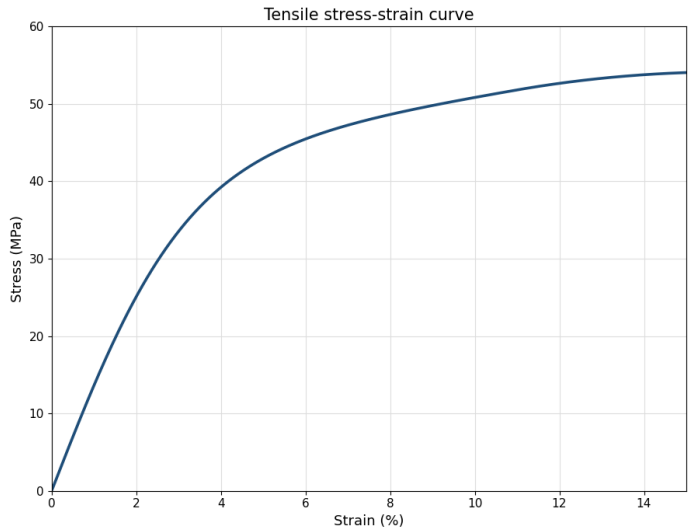


Figure 4.1: The sample figure: a single stress-strain curve, strain (%) from 0 to 15 on the X axis, stress (MPa) from 0 to 60 on the Y axis.

1 **Open the figure.** Help ▷ Open example ▷ Stress-strain curve. For your own figures: Open Image on the top bar, or drag the file onto the canvas.

2 **Capture the figure.** Frame it in the window and press **Capture figure**. What you see is frozen as the working image of record. The example opens with its graph type, **XY**, already picked. With your own figure, pick the type in the top bar: the figure is read in the background after capture, and the type it looks like is outlined green as **Suggested**.

3 **Detect the axes.** On the calibration card, press **Automatic axis detection**. It sets the data area to the plot's frame and places X1, X2, Y1 and Y2 on labelled tick marks, each with the number printed there: X 0 and 14, Y 0 and 60. Check them; move any point or retype any number if it is wrong.

4 **Calibrate.** Press **Calibrate**. The card folds to **Calibrated** ✓.

5 **Extract.** Open **Data extraction** (4) and press **Auto-extract all series**. This figure has no legend, so the series are the inks inside the data area: the curve is read as one line into **Series 1**, and the card reports what it found in one line.

6 **Sanity-check the result.** The curve should start at the origin and level off near **54 MPa** as strain approaches 15%.

7 **Export.** Export on the top bar, choose a format (CSV is the simplest for a first look), and save it or copy it to the clipboard.

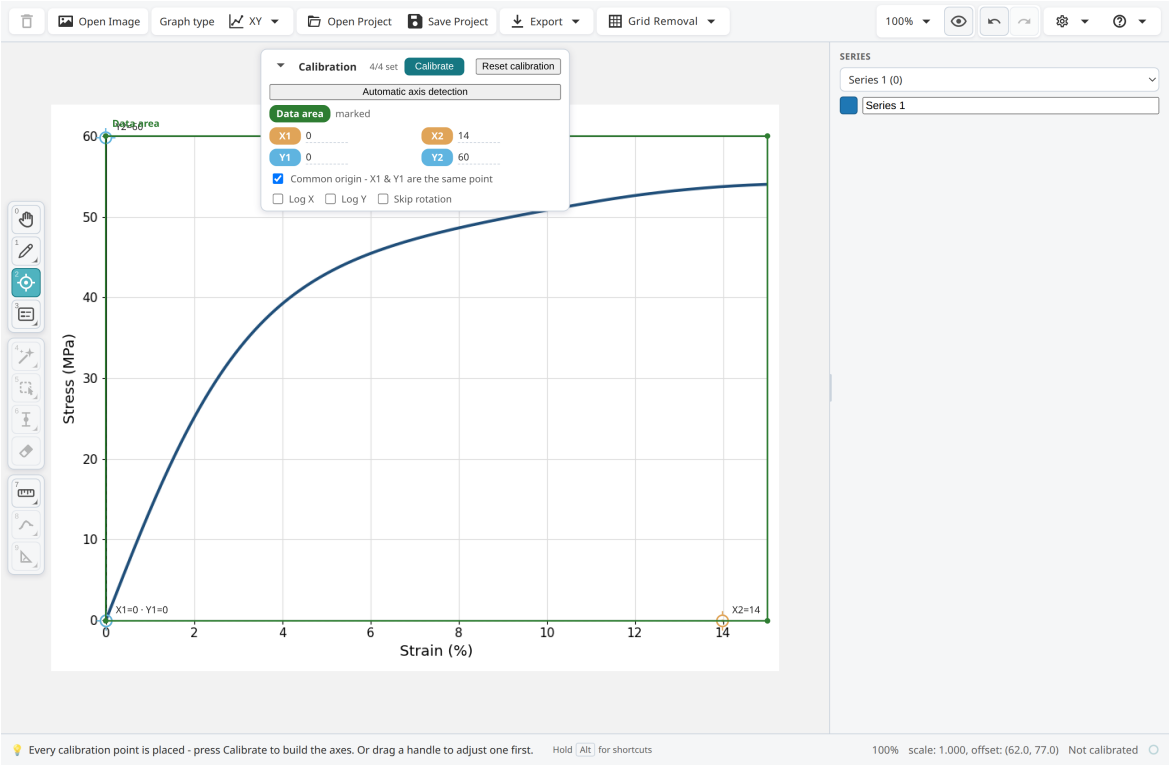
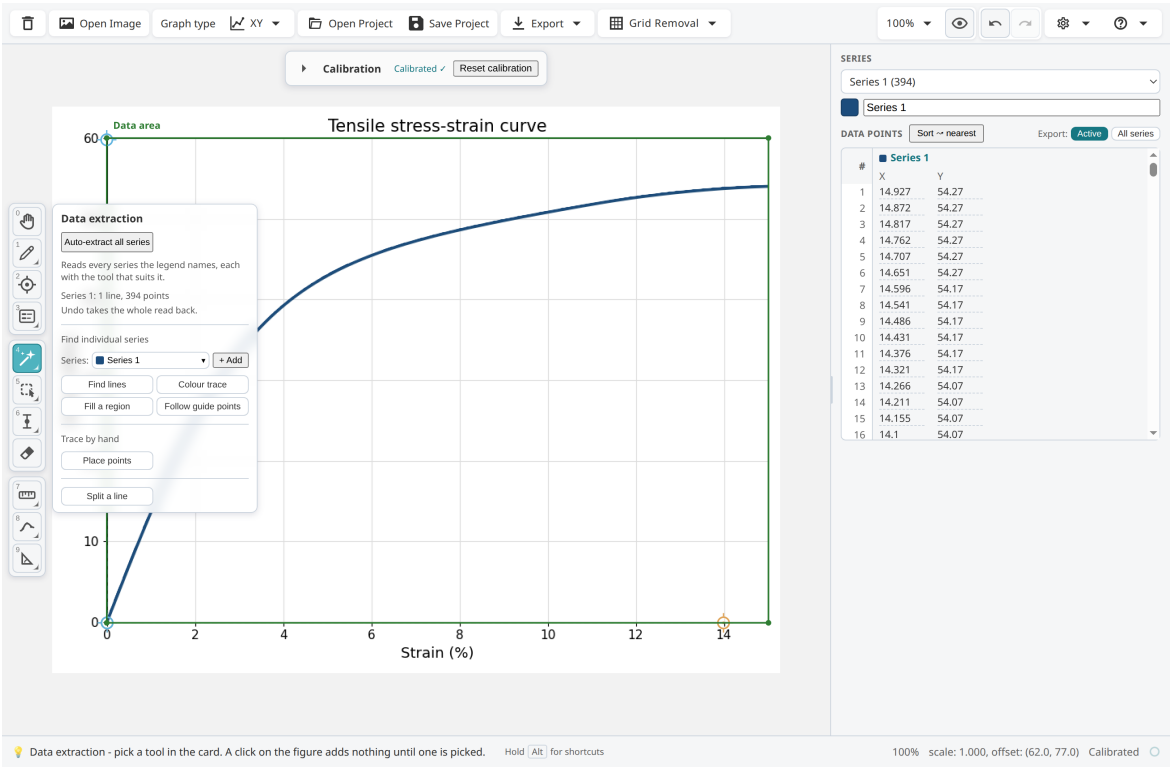


Figure 4.2: Step 3: Automatic axis detection has placed the data area and all four calibration points, each on a labelled tick with its number. Nothing is calibrated until you press Calibrate.

Every sample figure ships with a `.truth.json` file holding the figure's real calibration and data, useful for checking your own technique against a known answer while you learn the tool.

That is the whole loop the rest of this guide elaborates on: **open → capture → calibrate → mark the legend → extract → correct → export.** Each step is one card, and each card holds every way of doing that step: a click, the guided order, or an automatic detection.



#	X	Y
1	14.927	54.27
2	14.872	54.27
3	14.817	54.27
4	14.762	54.27
5	14.707	54.27
6	14.651	54.27
7	14.596	54.17
8	14.541	54.17
9	14.486	54.17
10	14.431	54.17
11	14.376	54.17
12	14.321	54.17
13	14.266	54.07
14	14.211	54.07
15	14.155	54.07
16	14.1	54.07

Figure 4.3: Step 5: Auto-extract has read the curve into Series 1, “1 line, 394 points”, and the table on the right holds the values.

Chapter 5

The Core Workflow

Each step below is one card, and each card holds every way of doing its step. The ways all make the same marks: any of them can finish or correct what another started, and nothing records which one was used.

5.1 Opening a figure

Open Image (top bar), drag an image onto the canvas, or paste from the clipboard (`Ctrl+V`). Supported formats: PNG, JPG, GIF, BMP, WebP, SVG, TIFF, and PDF (multi-page: you choose the page). Zoom with the scroll wheel (`Cmd/Ctrl`+scroll on a trackpad); pan with the middle mouse button or `Space`+drag; fit the view with `Ctrl+0`.

5.2 Capturing the figure

Frame the figure in the window (on a whole page, zoom and pan to it) and press **Capture figure**. This freezes what is framed as the working image of record, so the data always traces back to a stable source. With a PDF, the figure and its page are remembered in the project, and the calibration card says whether the figure is **drawn** in the PDF, a **picture**, or partly each. A drawn figure is read from its drawing, exactly, by the tools that can (Find lines, markers, bars, the axes' labels), and past 100% zoom it is redrawn sharp.

After capture, the figure is read in the background, and the graph type it looks like is outlined green in the picker, with **Suggested** under its name. It can suggest XY, Bar, Box plot, Pie, Map and Heatmap, and says nothing when it is not sure. Nothing changes until you pick a type yourself.

5.3 Preparing the image

Edit image (`1`) turns and trims the figure: **Rotate** and **Flip** in quarter turns, **Crop** to a dragged rectangle (the area to keep), and **Straighten** a scan by up to 15 degrees with the slider and **Apply**. **Auto-straighten** sets the angle from the X axis's calibration points once an XY chart is calibrated. Calibration points, data points and measurements move with the image, so no value changes.

Grid Removal (top bar) takes a colour and wipes every pixel within a tolerance of it, which helps when gridlines run through a curve. It knows nothing about charts, only about colour: check the figure afterwards, and undo if it took something it should not have.

5.4 Choosing the graph type

Pick the graph type from the picker in the top bar. Each type shows its own icon, so a bar chart and a histogram are told apart by shape rather than by two similar names. The picker is grouped by what the data *is*: Bar, Span chart and Stacked bar sit together because all three measure an extent against a baseline; the box plot and the candlestick sit under them because both name several values at one category. Chapter 6 covers each type.

5.5 Calibrating

The **Calibration** card opens after Capture figure, and clicking **Calibrate** on the rail unfolds it.

1. **The data area comes first.** Drag a box from one corner of the plot to the opposite corner. That is where the automatic tools search, so labels, titles and axes outside it are never read as data. It never stops a point you place by hand. (Measured: with no data area, Find lines took the labels, title and axes for lines on every bundled figure tried.)

2. **Then the calibration points, on labelled tick marks.** For XY, two X points and two Y points: click a labelled tick mark on the X axis and enter the number printed there, then a second one with a different number, and the same for Y. **Common origin** shares one point between X1 and Y1 when the corner where the axes meet is labelled on both.

3. **Press Calibrate.** The card folds to **Calibrated ✓**.

Automatic axis detection, just under the card's header, does that walk for you on XY, Contour, Histogram and the bar family. It sets the data area to the plot's frame and places every point it can read, each on a labelled tick with the number printed there. A drawn PDF figure is read from its own text and drawing; a picture by a chart-element detector and the text reader, offline. Check the points, move any of them or retype a number, place a missing one by clicking its chip and then the figure, and press **Calibrate**.

> **Any change to the calibration loses it**: a moved point, even a one-pixel nudge, an edited value, or an option such as Log X. **Calibrate** sets it again. While the figure is not calibrated, the data table keeps its rows and greys out the values.

A second stage: categories and grids

On the bar family and the heatmap, calibration has a second stage, and the card shows one stage at a time while you work. While you calibrate, a grey line names the stage that comes next ("Categories: next, after Calibrate"). Calibrate opens the card straight into it, and the card's heading carries the whole status: **Calibrated ✓**, the count ("4 categories", "5 × 5 cells"), the button that finishes the stage, and **Reset calibration**. Folding the card hides the stage's rows and leaves the heading.

A **category axis** is calibrated on two tick marks too: **Cat 1** where the first category's tick mark meets the axis line, **Cat n** where the last one's does, and how many categories the figure has. In the Categories stage you say whether the figure draws its ticks at the categories' centres (the default) or between them, and the marks move on screen to show which; **Find the figure's own ticks** moves

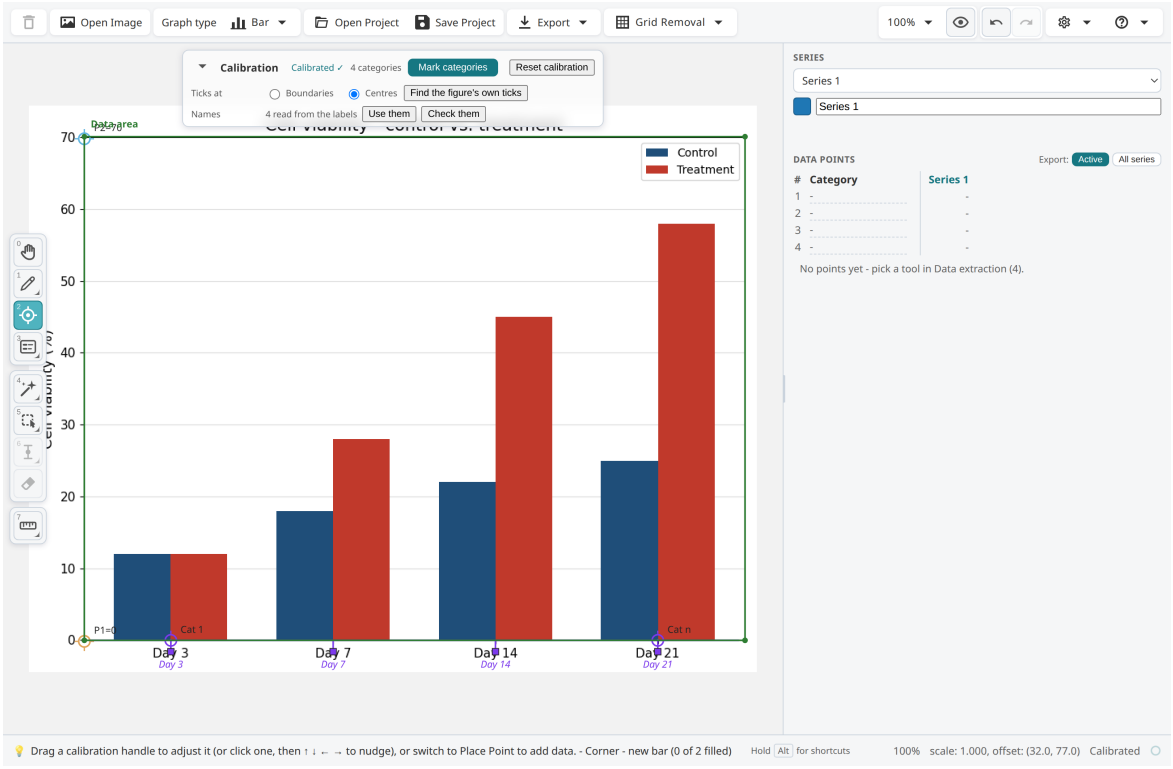


Figure 5.1: The Categories stage after Calibrate on a bar chart: the heading reads Calibrated ✓, 4 categories and Mark categories. The names read from the figure's labels are offered in italic under each tick mark, with Use them and Check them on the Names row.

them onto the tick marks the figure draws, and any tick can be dragged along the axis. On a bar chart, Calibrate also reads the category names from the labels and offers them: **Use them** writes them all in one Undo step, **Check them** opens them for correction first, and nothing is written until you do. **Mark categories** ends the stage and folds the card to **Calibrated ✓ Categories marked ✓**.

5.6 Marking the legend

Legend (3) has three buttons. **Mark legend**: drag a box round the figure's legend. **Auto detect legend** finds it for you, and says so in the tips bar when it finds nothing. **Mask an area**: drag a box over anything else that is not data, such as text, arrows, an inset or a reference line.

The legend and a masked area are rectangles laid over the figure, shaded in two colours: the picture stays as it was, every reader leaves them out, and their corners can be moved while the card is open. A marked legend is masked by itself. What is read from the legend, each entry's name and the sample beside it, names the series and tells the automatic extraction what to look for.

5.7 Extracting the data

Data extraction (4) is one card for every way to get points. When it opens, no tool is picked; pick the one that suits the figure. A tool that does not apply to the graph type is greyed, with the reason

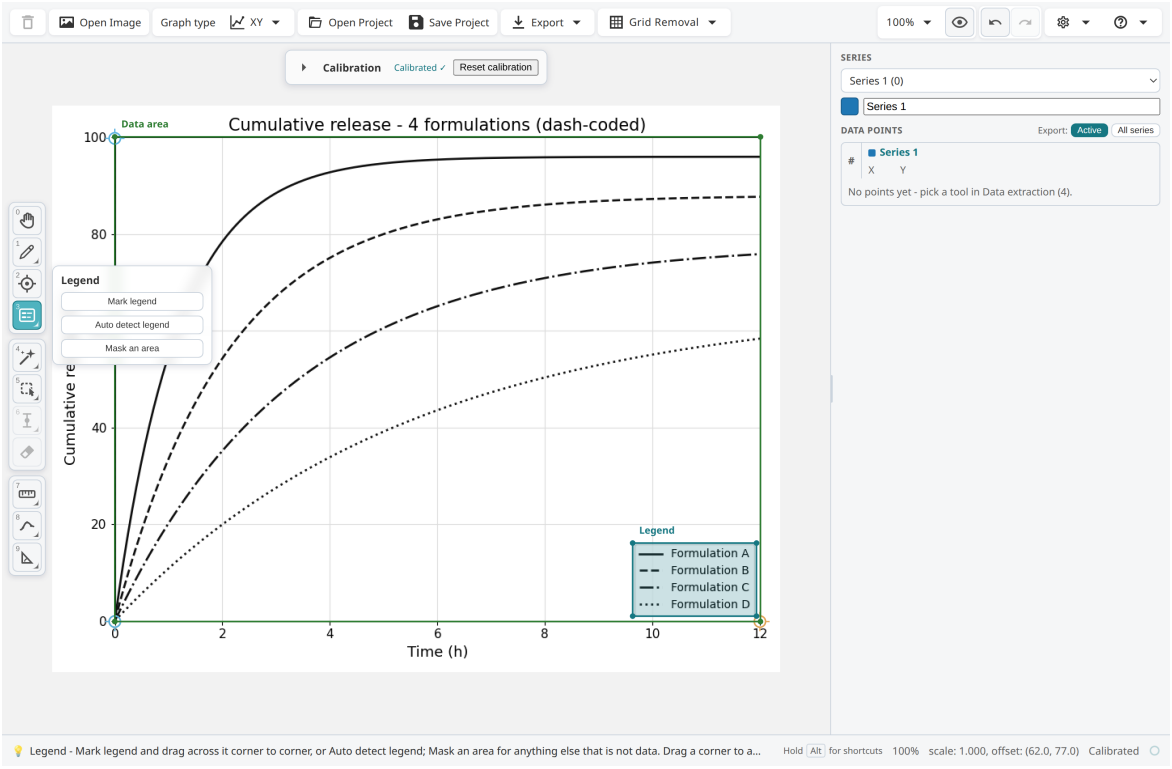


Figure 5.2: Auto detect legend has found the legend inside the plot. It is shaded over the figure and left out by every reader.

in its tooltip.

Auto-extract all series

Auto-extract all series reads every series the legend names, each with the tool that suits it, into a series of that name: a line in the legend with Find lines, a marker with Colour trace's scattered points, a bar with the type's own shapes.

- **Series in one ink are told apart by their line style.** The ink is read once, each line's pattern of dashes and gaps is measured and matched to the legend sample it looks like (solid, dashed, dash-dot, dotted or any other pattern), and each series' pieces are joined into one line.
- **Scatter series in one ink are told apart by their markers.** Each marker goes to the entry whose legend marker it looks like, and markers drawn over each other are taken apart into one point each.
- **With no legend marked**, the series are what the figure draws inside the data area: on a bar chart, each bar colour; on a line or scatter chart, each ink, read as a line where it is one long stroke and as points where it is many small markers, into Series 1, Series 2 and on, the most ink first. Ink that stays in one small corner (a legend inside the plot, an annotation) is not a series.

The card reports one short line per series. Every problem (a line no sample matches, a piece left beside its series' line, entries whose styles cannot be told apart) is listed once, under **Places to check**

on the right. One **Undo** takes the whole read back; pressed again, it replaces what it found before.

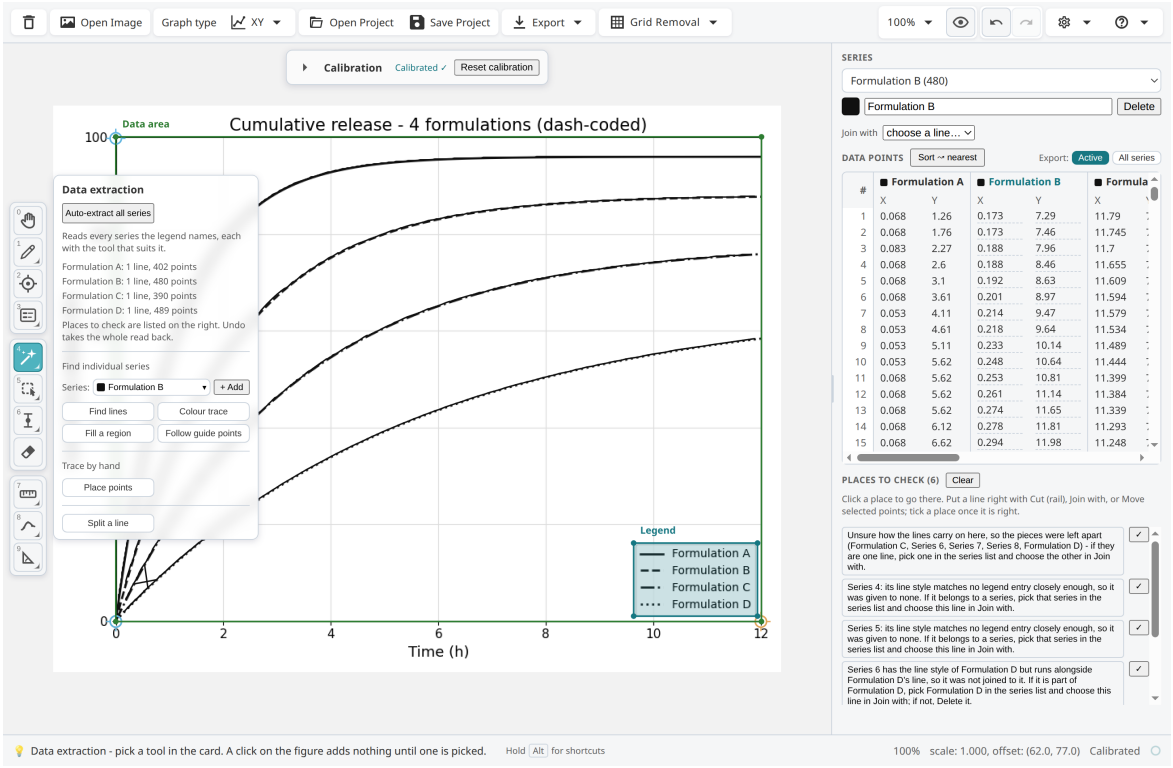


Figure 5.3: Auto-extract on four formulations drawn in one black ink: each is read as one line, named from the legend. The leftover pieces it could not place with confidence are listed under Places to check.

Finding individual series

The **Series** row picks the series to read into (rename it in place, or **+ Add** one). Then:

- **Find lines** (XY and Contour): every line of the picked colour becomes its own line, even where same-colour lines cross, touch, run together or are cut by their own labels. A gap the figure drew no ink in (a label, a dash, a crossing) is filled by a fair interpolation between the line's ends, marked `bridged`; a dotted line is read as a line; a stretch two curves share is recorded in both, marked `shared`; a spectrum's narrow peaks are read through their tips. Where it is unsure it does not guess: it leaves the pieces apart and lists the place under **Places to check**. **Find this line** takes just the line you click.
- **Colour trace**: every pixel of the picked colour inside the data area, in one pass. On XY and Contour it reads markers, one point per marker; on Polar, Ternary, Map and the circular chart recorder its Shape choice reads a curve or scattered points; on Bar, Histogram and Span chart it reads bars.
- **Fill a region**: click one point on a solid curve; it spreads along the connected ink inside the data area.
- **Follow guide points**: for curves that colour cannot separate. Click a few guide points along one line and a spline fills the rest. The guide points are your record; the fill is marked as derived.

- **Place points**: click along the curve by hand. A zoom loupe follows the cursor. This is the reliable default and works on any figure.
- **Split a line**: click a line where it should become two.

The colour Find lines and Colour trace read is shown at the bottom of the card, beside a small crop of the figure where it was picked, with **Pick from image**. Every automatic tool shows you what it captured, and one Undo takes it back.

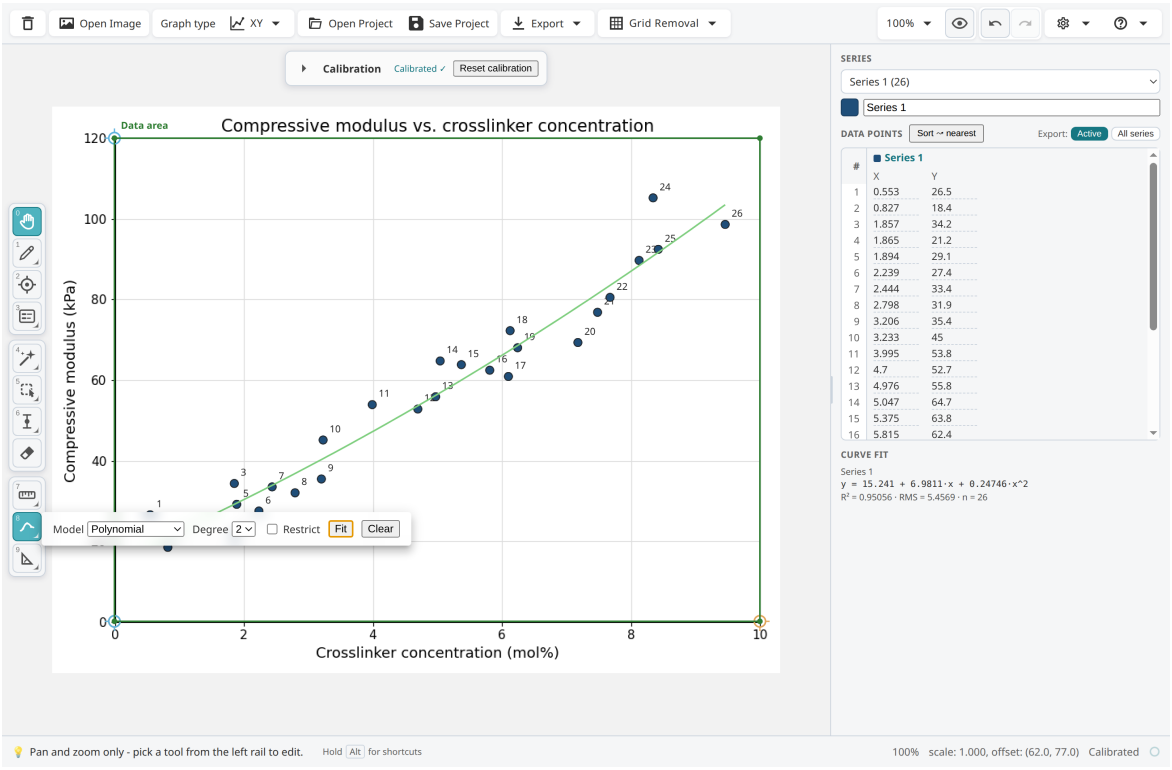


Figure 5.4: Scatter data read by Colour trace, one point per marker, with a curve fit drawn over it. The fit exports as its own block (§8.1).

5.8 Extracting by hand

Reach for this whenever the figure argues with you, and on anything where being right matters more than being quick. It works on every graph type and it never guesses.

1. Choose **Place points** on the Data extraction card. The loupe magnifies the pixels under the cursor, so you place each point against the ink rather than against your estimate of it.
2. Click along the curve. Each click is one reading, recorded where you put it.
3. Adjust without re-clicking: the arrow keys nudge the selected point, `Shift` makes the step coarse, `Q` and `W` step along the points, and `Del` removes one. The tips bar always says which of these apply.
4. Drag any point to move it. The number in the table follows, because the table reads the point back through the axes rather than storing what you clicked.

A value you type into the table is treated like a click: it moves the datum through the axes rather than overwriting the number, and it is shown in `[square brackets]` so a reader can see which readings came from your eye. That is not a lesser reading: there may be something in the colour, a hatch or a printed label that you can see and the sampler cannot.

5.9 Correcting

- **Places to check** (right panel) lists every place an automatic tool was unsure. Click one to go there; tick it once it is right.
- **Select** (`5`): click a point, or drag a box or a lasso to select a range, across every line inside it on XY and Contour. `Del` removes the selection; the arrow keys nudge it; `Esc` clears. It never selects calibration handles.
- **Move N selected to** (above the data table) moves the selected points onto another series, in order along it.
- **Join with** (series panel) joins the active line to another at their nearest ends; the gap is bridged and marked. Two lines with different levels are refused. Two curves swapped at a crossing: **Split a line** both there, then **Join with** the right pairs.
- **The Eraser** erases what it touches: click a point, or drag a box to erase every point inside it, in every series, on every graph type (a bar, a box or an error bar goes whole). A selection made with Select counts as one thing: one click on any of its points erases all of it, in one Undo.
- Edit any value in the right-panel table, on every graph type. Typing a number moves the datum to match it: on a spider chart it slides along that axis's own ray, on a heatmap it moves along the colour key, so the marker and the number can never disagree.
- **Undo** and **Redo** (`Ctrl+Z`, `Ctrl+Shift+Z`) cover everything, including image edits.

5.10 Multiple series and figures

+ **Add** on the Data extraction card starts another series on the same calibration, each named and colour-coded. **Extract another graph** re-enters a multi-page source (a paper's PDF, say) as a fresh figure; flip between figures with the arrows beside the calibration card. Each figure keeps its own image, calibration, series and graph type within the one project.

Chapter 6

Chart Types, One by One

Sixteen graph types: XY, Contour, Histogram, Line (categorical X), Bar, Span chart, Stacked bar, Box plot, Candlestick, Heatmap, Pie and donut, Polar, Spider or radar, Ternary, Map, and the circular chart recorder. This chapter covers what is particular to each; correcting, analysis and export are common to all of them.

6.1 XY (linear, log, date)

Two X references and two Y references, each axis linear, log or a date and time scale, chosen per axis on the calibration card. Automatic axis detection fills the walk. Chapter 4 is a full worked example.

6.2 Contour

A contour line is a line that carries a **level**: a number beside its points, not text in its name, so a plotting library can redraw the plot from the export alone. Several lines can share a level, and the table groups them under it. A line with no level exports blank, never 0.

- **Levels from labels.** Find lines reads each line's own label and offers it beside the line's Level: *The label reads 0.4*, with a Use button. It is offered rather than written in, because the reader gets some labels wrong, and a wrong level looks exactly like a right one.
- **Fill in the other levels** gives unset lines their level in equal steps from the ones you set, and says so instead when the levels are not equal steps.
- **Colour key.** Tick Colour key in the calibration and click the key's two opposite corners and two labelled ticks. Each line then takes its level from where its colour sits on the key.
- **Log level**, beside Log X and Log Y: filled-in levels follow it.

The export is one row per point: line, level, x, y, with `Level from` saying where each level came from (`typed`, `label`, `key` or `interpolated`).

6.3 Histogram

Calibrated like XY, Automatic axis detection included. A histogram bar's two corners are stored as genuine **bin edges and a height**, not a centre and a value, which matters for any statistics that depend on bin width.

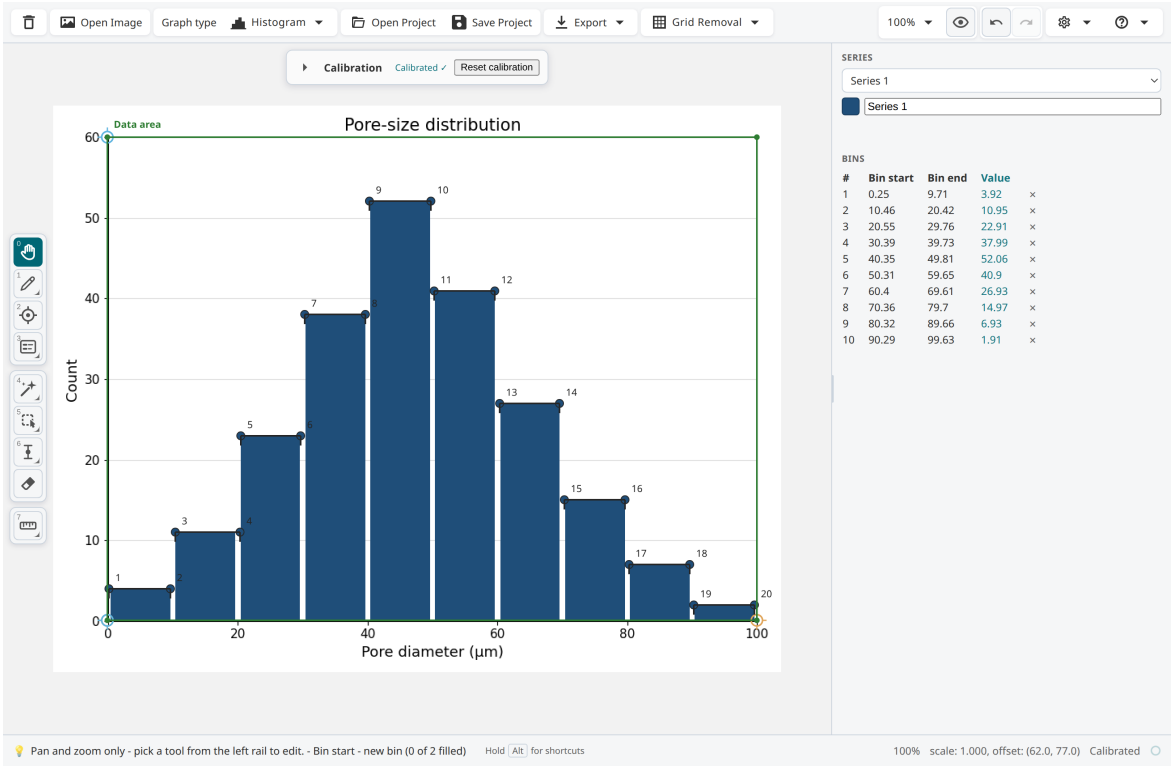


Figure 6.1: A histogram read by Colour trace: each bin's start, end and height.

6.4 Line (categorical X)

Behaves like XY for tracing and correcting, but the X axis is a category axis, calibrated on Cat 1, Cat n and the count. Each point carries its own **Category**, from the band it sits in.

6.5 Bar and column

Bars are captured as a drag-box. With **Place points**, press at one corner of a bar and release at the opposite one. A bar's value is one point, its end away from 0 on the value axis; a bar below 0 reads negative. A bar that floats clear of the axis belongs on a **Span chart**: on Bar it reports a note rather than a plain reading.

Bars are found whole by the chart-element detector. On a picture, Colour trace and Auto-extract ask the detector where the bars are. Where it is sure of a bar, its box is one bar, placed once with the detector's edges and filed under the colour that fills most of it; a hatched bar, which a colour read would find in pieces, is placed whole. Where the detector is sure of no bar, the bars are read by colour alone. The first press on a figure loads the detector and can take a few seconds.

The table is **one row per category and one column per series**, so a grouped figure shows every series at once. Every bar is filed by the category it stands in, because the categories are part of the calibration: the readings line up whatever order you capture them in, and if two bars of one series land in the same category, the table says so underneath.

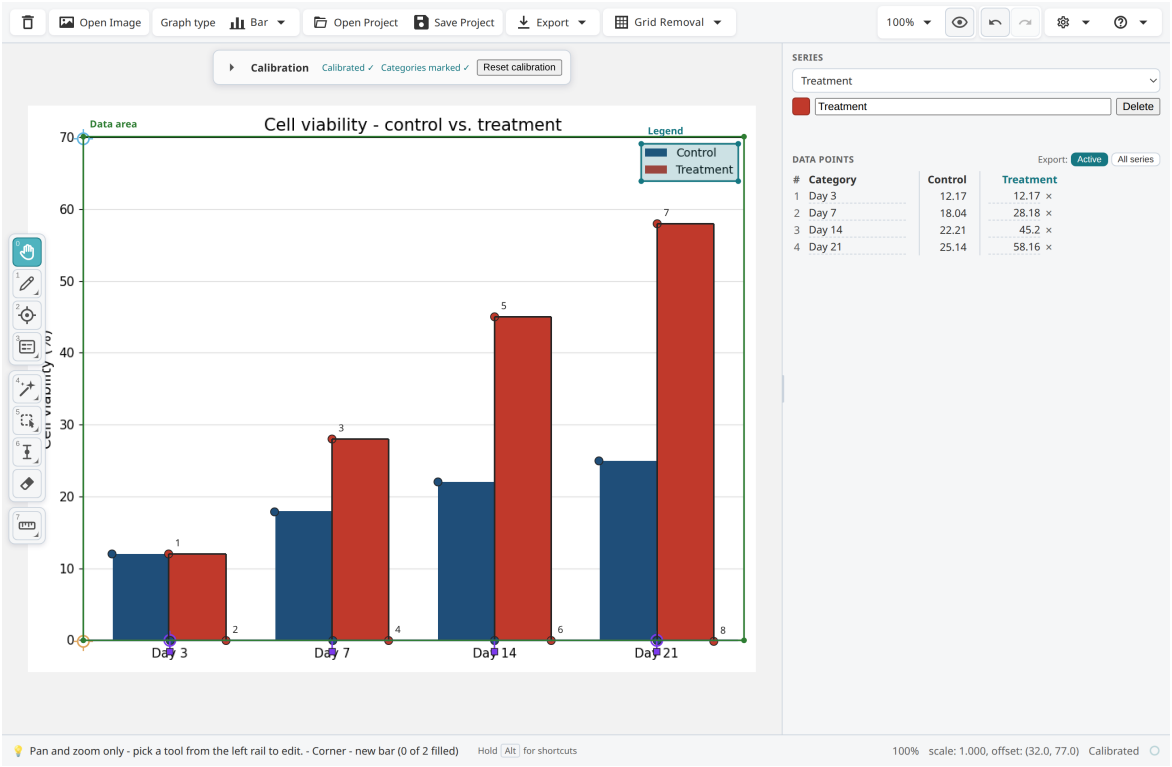


Figure 6.2: A grouped bar chart read the automatic way: axes and categories detected, the offered names used, the legend detected, and both series read by Auto-extract, one row per category and one column per series.

Names are never invented. An unnamed category reads as a dash, not as "Bar 1": a placeholder in an export's Category column is indistinguishable from a name someone transcribed. Use the names detection offers, read them with **Read labels from the figure**, or type each one once, in its row.

6.6 Span chart

A bar that floats clear of the axis (a temperature range, a tornado chart) is a *span*: both ends are measurements, where a bar has one measurement and a reference line. Capture it with the same drag-box; the two ends are reported as `Min` and `Max`, sorted by value rather than by which corner you pressed first. Each end can carry its own error bar.

6.7 Stacked bar

A stacked bar chart is its own type because its segments are *connected*. A segment's bottom is never a number of its own: it is the top of the segment below, and the lowest segment sits on the baseline. So the reported `Value` is a **height**: a segment running from 2 to 5 reads 3, with its `Base` of 2 shown beside it. Type a number into `Value` and the segments above move up and keep their own values. Dragging a segment's top does the same; the other corner measures the bar's width and changes no

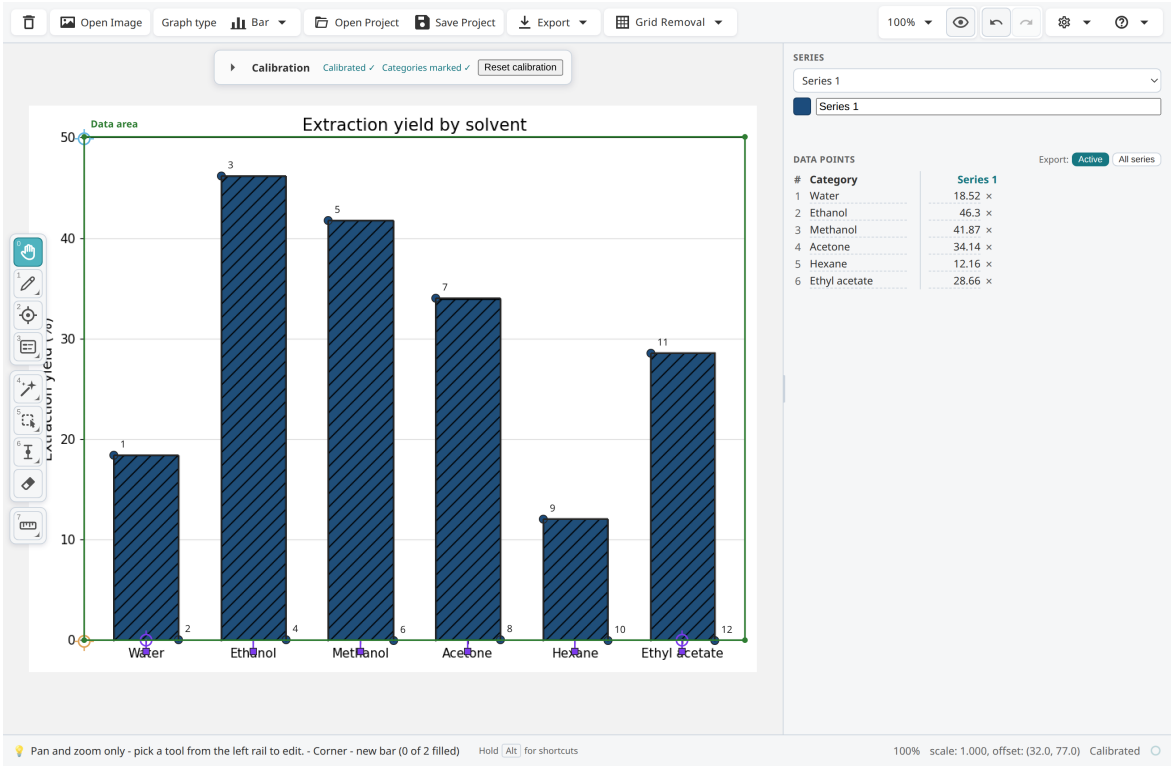


Figure 6.3: Hatched bars with no legend: Auto-extract reads the bar colour, and the detector gives each hatched bar whole. The category names came from the labels.

value.

Order comes from the figure: the segment nearest the baseline is the first, and each one after it is the next edge outward, so a stack drawn downward from zero needs nothing switched on. A column floating clear of the baseline is a different figure, which PlotTracer does not read yet.

6.8 Box plot

Five marks per category, clicked from the bottom up: the minimum, the lower quartile, the median, the upper quartile and the maximum. Colour trace is greyed here by design: a box's five values are not a bounding box, and a plausible box would misread the figure. The values are exported side by side, named the same way as Bar categories.

6.9 Candlestick

Four marks per period, clicked from the bottom up: the low, the lower body edge, the upper body edge, the high. The prompts name *places* rather than open and close, because on a falling candle the open is the top edge. Which edge is the open is read off the figure: each body's colour is sampled and the candles are grouped into the two appearances the figure draws. A card at the bottom of the canvas says what was found, and **Swap which colour rises** changes every candle at once. A figure drawn in one colour has no direction to read, so every candle is reported rising.

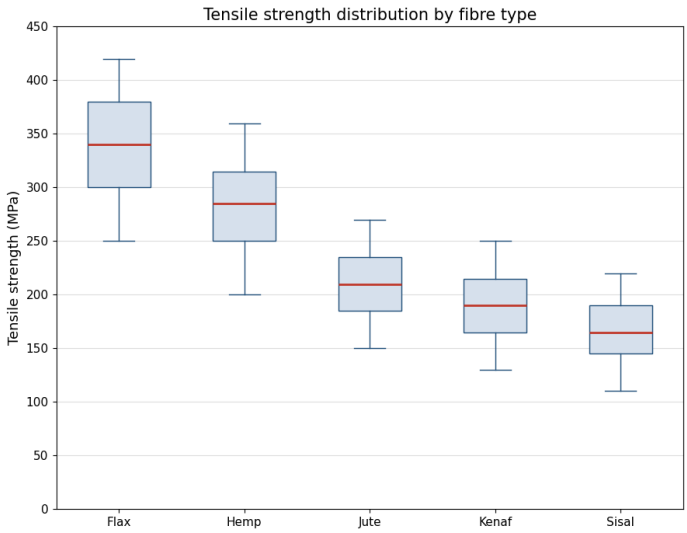


Figure 6.4: A box plot, one of the bundled samples.

6.10 Heatmap

A heatmap is a matrix: two banded axes and a colour axis, and the colour *is* the value. You calibrate the two axes and the colour key, put a grid over the cells, and every cell is read through the key.

- **Three corners describe the frame**: first column × first row, last column × first row, first column × last row, with the column and row counts. Three points describe a rotated or skewed figure as happily as a straight one; three in a line are refused.
- **Each axis is Values or Categories**, declared before you click. A value axis asks for the coordinate at each end; a category axis asks only for its count, and whether its marks sit between the bands or under them.
- **The colour key**: two clicks for where the strip begins and ends, then two printed ticks with their numbers. Tick Log colour scale for a logarithmic key.
- **The Grid stage**: **Detect grid** proposes boundaries from the figure's own ink, and says so and places nothing when it finds fewer than it needs; **Overlay even grid** lays one for a continuous field. Every boundary can be dragged, added or removed. **Read column names** and **Read row names** read a named axis's labels.
- **Read cells**, on the card's heading, reads every cell and folds the card to the number of cells read.

Each cell reports its own evidence beside its number: the range of values its colour cannot be told apart from, how far its colour sat off the key, how much of the cell was that colour, and whether it sits at the key's limit, where the figure may have clipped the value. Pick a cell and type the number you can see to correct it: the cell moves along the colour key, and the export names its source.

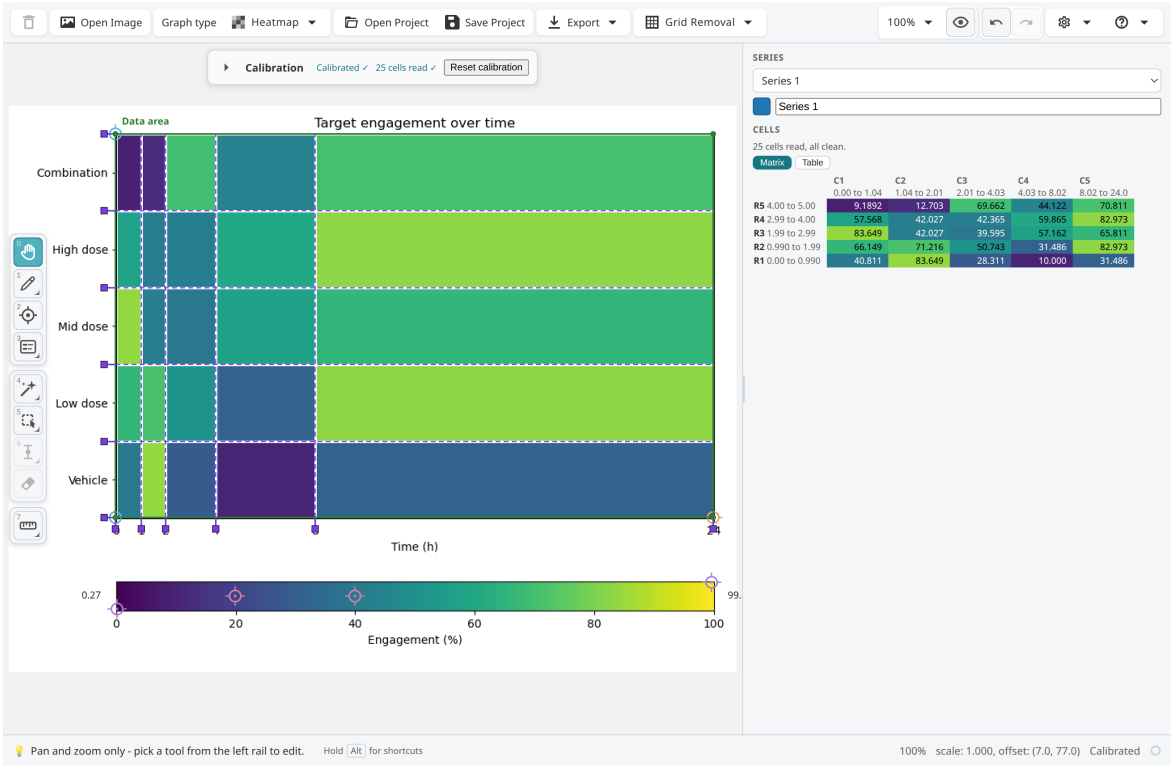


Figure 6.5: A heatmap with unequal columns, read through its colour key: the matrix on the right is tinted with the figure's own colours.

A radial (polar) heatmap cannot be read, and PlotTracer cannot tell one from a rectangular figure: the calibration builds and every cell reports a confident number that means nothing. A key drawn as a few discrete bands is refused, and says why.

6.11 Pie and donut

Calibrated from the **outline**, not the centre. Click three or more points round the rim: three fit a circle exactly, five or more fit an ellipse and report how well the rim really is one. The centre is worked out from the fit; a donut has no centre to click.

- **Total** (100): what the whole circle is worth. Leave it and slices read as percentages; type the figure's own total and they read in its units.
- **Sweep** (360): how much of a turn the chart draws. A half-pie or a gauge has a smaller sweep.
- **Tilted / 3D pie** for a chart drawn in perspective: read flat, a tilted 7% slice can read 13%, and the slices still sum to 100.

Capture one click per boundary: slices share their edges, so a ten-slice pie is ten clicks. When you come back round, the first boundary offers to close the ring; nothing closes it for you, because a half-pie should not close. An **exploded slice** uses the Exploded slice button at the bottom right of

the canvas: click the slice's own tip, then its two edges, and the reading is taken about that tip.

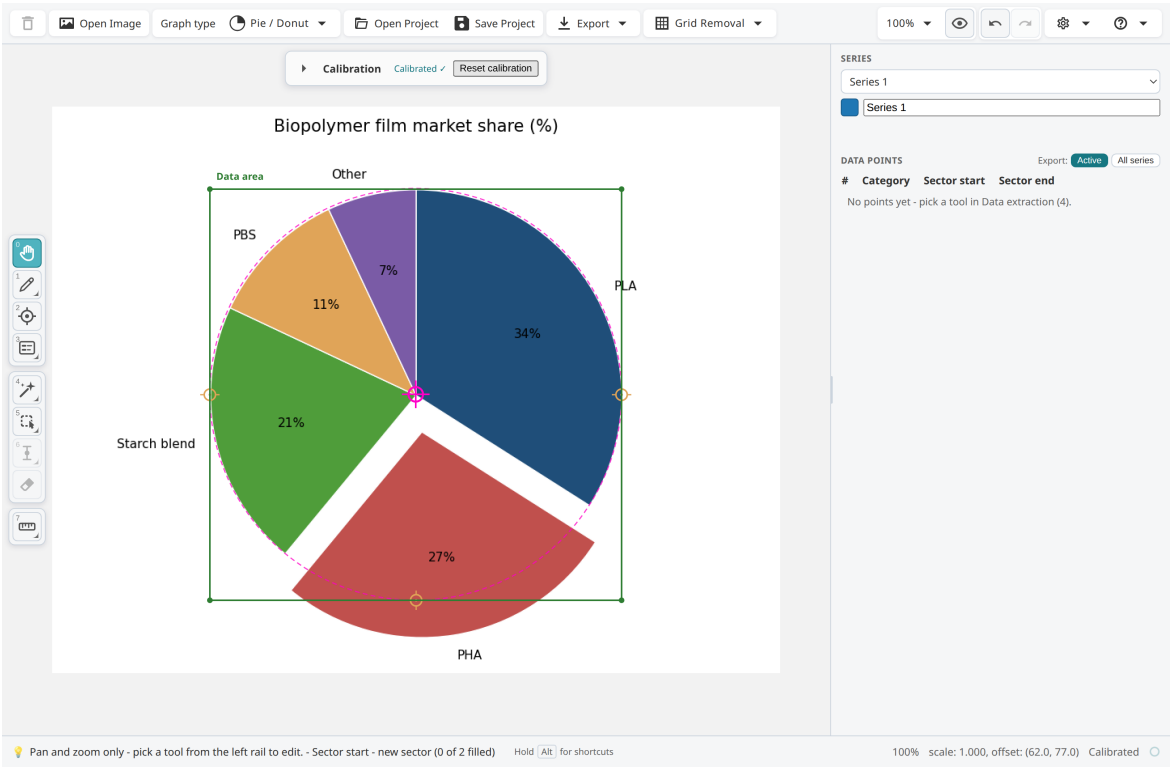


Figure 6.6: A pie calibrated from its outline: the fitted rim and centre are drawn back over the figure, an exploded slice included.

6.12 Polar

The card asks one question first: **Shape**, Circular or Tilted or squashed. For an ordinary polar plot, leave it on Circular: click the centre (r = 0), then a point with known r and θ, then a second point with a known r along the same angle, and give each its value. **Direction** says which way the angle runs, and **Log radial** is an option.

Choose **Tilted or squashed** when the rings are ellipses: the walk then asks for the second point's angle as well, at a different angle from the first, and the radial value at the centre. Two points at different angles measure the frame the figure was drawn in, so radii and angles are read through its own shape. On a test figure squashed to 60% of its height with a shear, readings came back within about 1% of the true values.

6.13 Spider / radar

The number of axes belongs to the figure. Click the **centre** and give the value every axis starts from (0 unless the chart says otherwise). Then, for each spoke, click one point of known value on it and type that value and the axis's name: one click gives both the ray's direction and its scale. **+ Add axis** adds as many as the chart draws. Each axis keeps its **own range**.

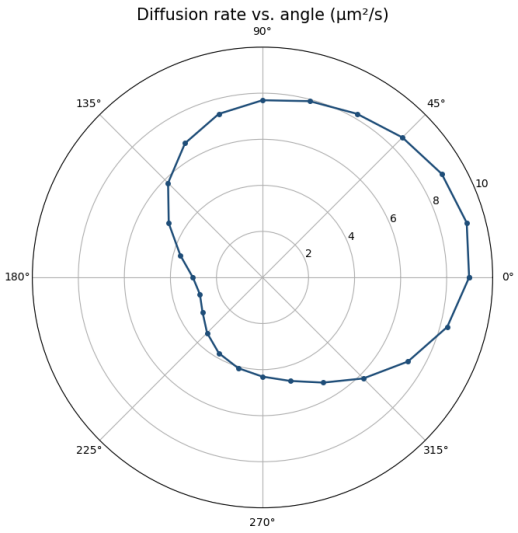


Figure 6.7: A polar figure, one of the bundled samples.

Capturing steps round the axes: the live one is drawn in magenta, and your click is recorded where it crosses that ray. **Colour trace** walks each ray outward and records where the series' colour crosses it. Where a ray crosses the colour more than once, it records **nothing** on that axis and names it, and the capture cursor lands there next, so what it refused is what you are asked for.

6.14 Ternary

Click the three corners, going round the triangle, and name each as the figure does (`Sand`, `Silt`, `Clay`); without names the components read A, B and C. Any click order reads correctly, and a triangle drawn with a right angle reads correctly too, because the reading is the point's composition within the triangle you clicked. All three components are stored, and they sum to the range you chose by construction. Three corners in a line are refused at the click.

6.15 Map

A map is calibrated on a **scale bar**: click one end of a reference line of known real-world length, then the other end, and type its length. **Units** names the unit (km, say), and **Origin** puts the zero at the bottom left or the top left of the figure. Points are then read as distances in those units.

6.16 Circular chart recorder

The disc charts of analogue loggers (temperature, pressure, humidity), whose time runs round the disc and whose value runs outward. Say how long one **Rotation** is (a week or a day) and its **Direction**, then walk five points: three on the pen's time axis at a known low, middle and high value (`(T0,R0)`, `(T0,R1)`, `(T0,R2)`), then two more at the high value at other times, as far apart as possible. **Chart Start Time** sets the clock the times are read against.

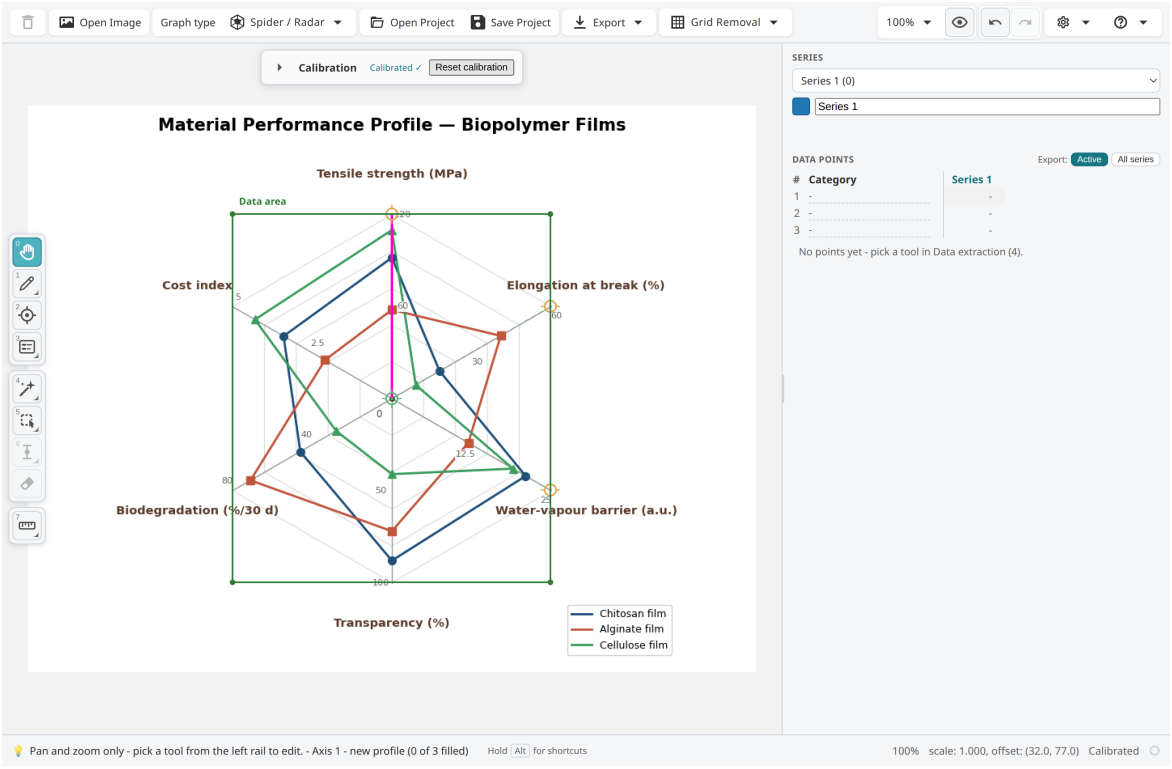


Figure 6.8: A six-axis spider chart calibrated, each spoke on its own scale.

6.17 Error bars

Error bars are **not** a graph type: they are a rail tool (6), captured on top of the series they belong to.

1. **Set up.** Error for: the series the bars belong to (it needs a point already). Name: what the uncertainty is called on the figure, such as `SD`, `95% CI` or `SEM`. That name is the only meaning PlotTracer records.

2. **Capture.** Drag from a data point out to its cap. The start snaps onto the point; the cap end is never snapped, because that end is the measurement. Up or down records `upper` and `lower`; left or right records `left` and `right`.

3. **Move the mirrored cap.** One drag places both caps: the one you dragged, and its reflection as a starting position. On an asymmetric figure the reflection is wrong; drag it to where the figure draws it. A cap slides only along its own bar.

On a Span chart each end carries its own error. A box plot and a candlestick take none: their values already are the spread. The caps export as columns beside their value, with both the absolute positions and the ± delta, so matplotlib's `yerr` and ggplot's `ymin`/`ymax` both have what they need.

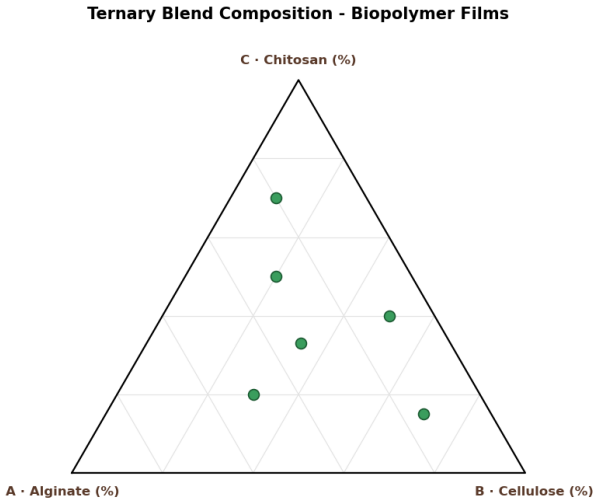


Figure 6.9: A ternary composition diagram, one of the bundled samples.

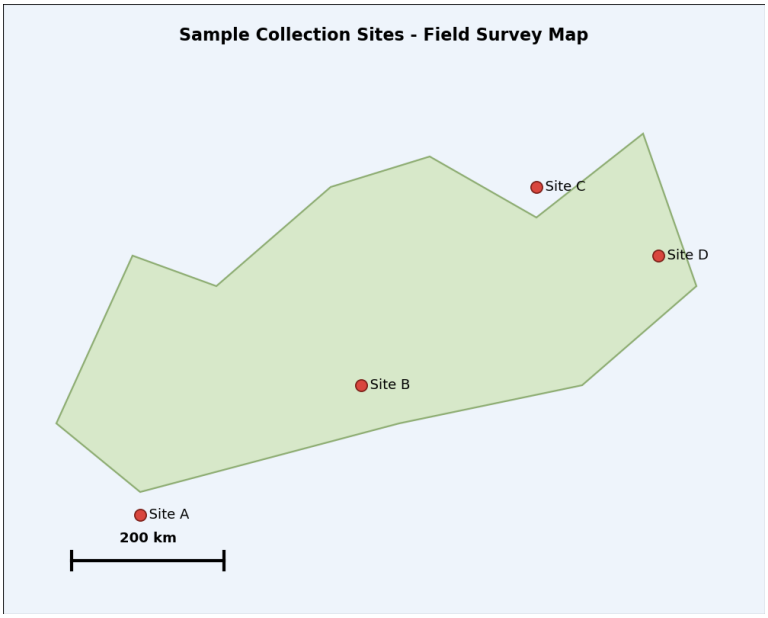


Figure 6.10: A map with a scale bar, one of the bundled samples.

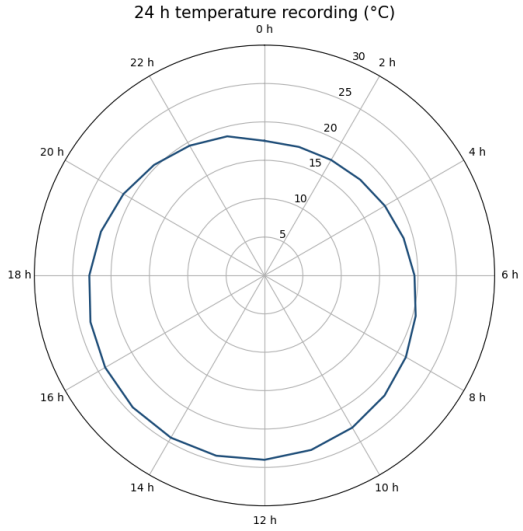


Figure 6.11: A circular chart recorder trace, one of the bundled samples.

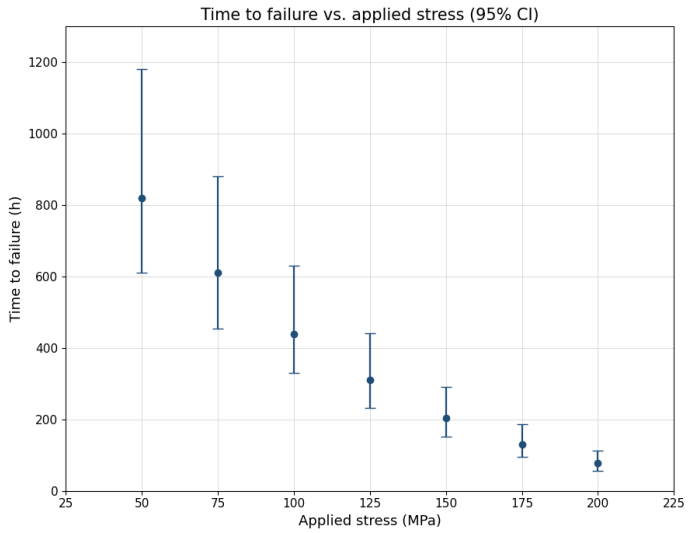


Figure 6.12: Asymmetric 95% confidence intervals, one of the bundled samples: every mirrored cap here has to be moved.

Chapter 7

Measuring

The ruler tool (7) opens a Measure card with four modes: **distance**, **angle**, **area** and **slope**, read in the chart's own units or in a scale you set from any known length on the image. Measurements are kept apart from your series, so they never mix into an export of the figure's points.

Chapter 8

Analysis

8.1 Curve fitting

Curve fit (`8`) fits a model through a traced series.

Model	Form
Polynomial	$y = a_0 + a_1 x + a_2 x^2 + \dots$ (choose the degree)
Exponential	$y = a \cdot e^{bx}$
Power	$y = a \cdot x^b$
Logarithmic	$y = a + b \ln x$
Gaussian	$y = a \cdot e^{-(x-b)^2/2c^2}$
Logistic	$y = a/(1 + e^{-b(x-c)})$

The card reports the equation, its coefficients, R^2 and RMS, and draws the fitted curve over the figure, only where it was fitted. Some models cannot take all data (a logarithmic fit needs every x above zero, a power law the same, an exponential every y above zero), and PlotTracer says which requirement is unmet rather than returning a number it cannot stand behind. Use **Restrict** to fit over a chosen x range.

> **A fit that did not settle says so.** The nonlinear models are solved iteratively (Levenberg-Marquardt), and a solver that runs out of iterations still returns *something*. Then the card says the curve is where the solver stopped, and the exported `Curve fit` block carries `settled = no`. A polynomial is solved directly and reports `n/a`.

The fit is always exported as a **separate block**: the points you placed are never overwritten by the model drawn through them.

8.2 Geometry and statistics

Geometry (`9`) gives arc length, enclosed area and curvature, computed from a traced series or a fitted model.

Chapter 9

Exporting Your Data

9.1 Formats

Export on the top bar offers CSV, TSV, JSON, OpenDocument (`.ods`), Excel (`.xlsx`), LaTeX, MATLAB, Python, R, or a PNG of the annotated figure. Any text format can also be copied to the clipboard. Choose **Active** (the current series) or **All series**.

Values are rounded to the figure's real resolution, never padded with false precision and never collapsed to zero, with a full-precision option when every digit is wanted. Fitted curves, geometry and measurements export as their own blocks, apart from the recorded points.

> The format menu says what each format *will not* carry before you pick one. No data format carries the figure, the calibration or the source document: save a **project** to keep those.

9.2 The role column

A series with derived points gains a **role** column: `anchor` for a guide point placed by eye, `interpolated` for one the spline filled in, `bridged` for a point Find lines filled across a gap, `shared` for a stretch two curves run along together (recorded in both), and blank for an ordinary point. Keep only the blank and anchor rows for strictly what was read off the figure. The column appears only where a series has such points.

9.3 Project files

Save Project writes a self-contained `.zip` holding every image, calibration, series and measurement, and the original source PDF, so the whole extraction reopens exactly and any number traces back to its figure.

9.4 Reading other tools' projects

Open Project also reads projects written by other digitisers: WebPlotDigitizer `.tar` archives, Engauge Digitizer `.dig` files and StarryDigitizer `.zip` projects. The format is recognised from the file's own bytes, never its name, and a file no filter recognises is refused with the list of formats that do work. A project holding several figures lists them all and asks which to open.

This is a **one-way** road. PlotTracer reads those formats but does not write them: a third of what it records (spider spokes and their per-axis scales, point roles, box-plot values, measurement blocks) has nowhere to go in any of them.

Chapter 10

The Trace Challenge

Help ▷ Take The Trace Challenge is a game: four figures (two easy, one medium, one hard), each already calibrated, traced against the clock. Read each round's instruction, place the points, and press **Done**. Every point missed or placed off its true value adds time, and the lowest total time wins.

It is traced **by hand**: it tests your eye and your aim. During a round the Data extraction card offers **Place points** (and **Split a line** where lines are the record); the automatic tools stay in their places, greyed, saying *"The Trace Challenge is traced by hand."*

Chapter 11

Troubleshooting

Automatic axis detection left a point empty, or put one in the wrong place.
It places only what it could read. Click the point's chip, then the labelled tick on the figure, and type the number printed there; or drag a misplaced point and retype its number. Then press Calibrate.

Auto-extract read one series where the figure has several.
With no legend marked, series drawn in one ink come out as one series. Mark the legend (or press Auto detect legend) and run Auto-extract again: it then tells them apart by line style or marker.

A legend inside the plot was read as data.
Mark it with Mark legend, or cover anything else that is not data with Mask an area. Every reader leaves both out.

A spider chart's spokes disappeared after Grid Removal.
Grid Removal strips every pixel near the sampled colour, whatever chart it is. If the figure draws its rings and spokes in the same grey, both go. Undo, and trace before removing the grid.

Some spider axes are blank after Colour trace.
By design: where a ray crosses the colour more than once, the right crossing is in doubt, so nothing is recorded and the axis is named. Place that reading by hand.

A curve fit looks wrong, or says it did not settle.
Restrict the fit to a narrower x range, or check the traced points are clean before fitting again.

An exported category is blank instead of named.
Deliberate: PlotTracer never invents a name like "Bar 1". Use the names detection offers, read them with Read labels from the figure, or type them from the figure.

macOS says the app is damaged, or from an unidentified developer.
The builds are unsigned. Right-click the app and choose *Open*; this is needed once per install.

Windows SmartScreen blocked the installer.
Click *More info*, then *Run anyway*: an unsigned installer, not a damaged download.

A project written by another digitiser will not open.
Only WebPlotDigitizer, Engauge Digitizer and StarryDigitizer projects are recognised (§9.4), by the file's own bytes. A file from one of those tools that still is not recognised may be a version the reader does not cover yet.

Chapter 12

About This Project

12.1 License

PlotTracer is distributed under the **GNU Affero General Public License v3.0**, inherited from WebPlotDigitizer upstream and kept in every distribution.

12.2 Attribution

PlotTracer's calibration engine began as a TypeScript port of **WebPlotDigitizer** by Ankit Rohatgi (AGPL-3.0); the application itself (interface, interaction model, workflow) is a ground-up rebuild. Several algorithms (flood-fill curve tracing, grid-line removal, curve fitting, geometry and statistics) are clean-room reimplementations of ideas from **Engauge Digitizer** (Mark Mitchell, Jason Nicholson; GPL-2.0), written from the algorithm descriptions rather than translated from source. PlotTracer reads **StarryDigitizer** projects (MATO Tomoya; MIT) at the file level. Text is read with the **PP-OCRv6** models from **PaddleOCR** (Apache-2.0, converted to ONNX), run by **ONNX Runtime** (MIT). The chart-element detector's model card ships with the application and names what it was trained on.

12.3 No telemetry

PlotTracer makes no network calls of its own. Your figures, your data and your project files never leave your machine unless you choose to share them.

12.4 Getting help

Issues and pull requests are welcome on the project repository: `https://github.com/katalystnord/plottracer`.